

End of Topic Test – Chapter 4: Critical Analysis (Test 2)

Answer all questions in the spaces provided.

1. Identifying evidence, reasoning and conclusions

Read the following extract:

“Speed cameras should be installed along the ring road. A recent survey of 350 local drivers found that 74% said they would reduce their speed if cameras were installed. This proves that installing speed cameras will reduce the number of accidents on the ring road.”

- (a) (2 points) Identify the *evidence* used in the extract.

- (b) (2 points) Identify the *reasoning* used.

- (c) (2 points) State the *conclusion* of the extract.

- (d) (3 points) Criticise the argument, referring to sample representativeness, assumptions, or logical leaps.

2. Evaluating claims about breakfast clubs and attainment

A local newspaper claims:

“Free breakfast clubs in primary schools will significantly improve pupils’ exam results.”

The Department for Education recently stated:

“Over 2,700 primary schools now offer a funded breakfast club... The programme aims to help pupils arrive at school ready to learn and to improve attendance and concentration.”

A review by the National Pupil Research Trust (NPRT) notes:

“Evidence on the impact of breakfast provision on attainment is mixed. Several studies report small improvements in concentration and attendance, but effects on exam grades are inconsistent and often not statistically significant.”

- (a) (3 points) Explain how the Department for Education quotation supports the newspaper’s claim.

- (b) (4 points) Explain how the NPRT review quotation challenges or weakens the newspaper’s claim.

- (c) (3 points) Evaluate the overall strength of the newspaper’s claim, using both quotations.

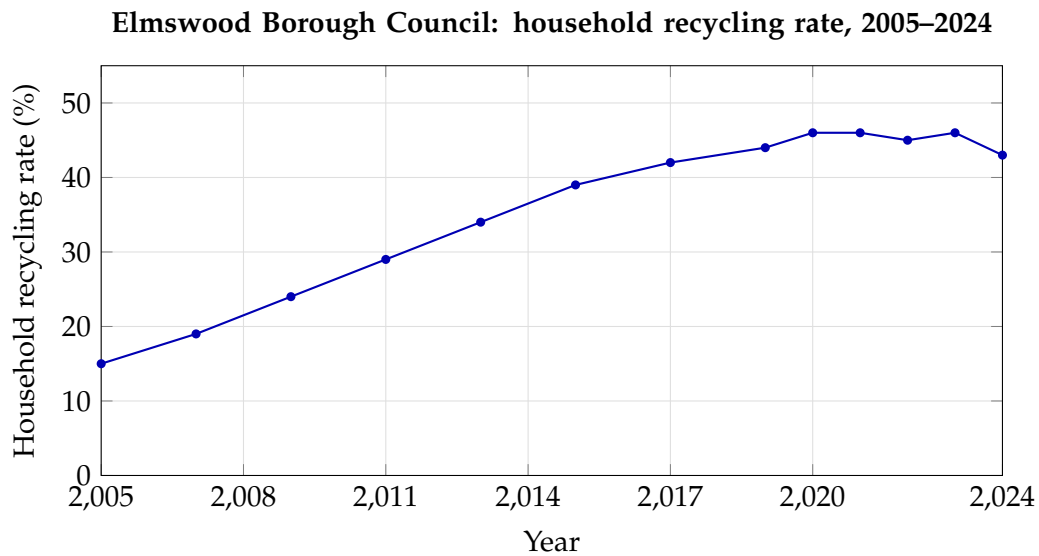


Figure 1: Household recycling rate in Elmswood, 2005–2024 (illustrative data).

3. Interpreting claims about recycling trends

A local newspaper recently ran the headline “Elmswood’s Recycling Drive Is Collapsing”, after the council’s household recycling rate fell from 46% to 43% between 2023 and 2024. It reported:

“Recycling rates in Elmswood fell for the first time in over a decade last year, undoing years of progress and prompting warnings that the council’s recycling drive is now collapsing.”

The council’s environment officer responded:

“Long-term figures show recycling in Elmswood has more than tripled since 2005, and the recent dip follows a decade of steady improvement.”

The chart above shows the long-term recycling trend in Elmswood.

- (a) (3 points) Give three reasons why the headline claim that the recycling drive is “collapsing” may be misleading when compared with the long-term trend shown in the chart.
- (b) (3 points) Explain how the environment officer’s quotation challenges the idea that Elmswood’s recycling drive is collapsing.

- (c) (3 points) Suggest one additional piece of information that would help a reader judge whether Elmswood's recycling scheme is genuinely failing.

4. Sampling and bias

A gym wants to find out whether its members would be willing to pay a higher monthly fee for new equipment. The gym manager asks the 25 members who attend the 6 a.m. spin class on a Monday morning.

- (a) (2 points) Identify the sampling method used.
- (b) (3 points) Give reasons why this sample may be biased.
- (c) (3 points) Suggest a more appropriate sampling method and explain why it would be better.

5. Correlation and causation

A study finds a strong correlation between monthly ice cream sales and the number of recorded drowning incidents at UK beaches. A researcher summarises the findings:

“Data collected over several years shows that as ice cream sales rise, the number of drowning incidents also rises. Months with the highest ice cream sales consistently show the highest number of drownings, and months with the lowest sales show the fewest incidents.”

- (a) (2 points) State the meaning of “correlation” in this context.
- (b) (3 points) Explain why the correlation does not necessarily mean that ice cream sales cause drownings.
- (c) (2 points) Suggest one additional piece of data or research that could help investigate the true cause of the increase in drownings.

6. Reasoning flaws: meal kit boxes and food waste

A newspaper article begins “Are Meal Kit Boxes Solving Britain’s Food Waste Problem?”:

“Meal kit companies deliver pre-measured ingredients for each recipe, so customers only buy exactly what they need. This fact drives their popularity as a low-waste way to cook. Subscriber numbers for meal kit services have grown by over 18% each year since 2021, fuelled by growing concern about food waste and a desire for convenient cooking.”

However, a sustainability review notes:

“Meal kit boxes do not automatically deliver the low-waste outcome their marketing suggests. Each box typically contains significant packaging – individual plastic wrapping, ice packs and cardboard – much of which is not recycled. Their overall environmental impact depends heavily on packaging materials, delivery distances and whether customers actually cook every meal provided.”

- (a) (3 points) Critically analyse the newspaper’s reasoning.

- (b) (3 points) Suggest two alternative explanations for the rise in meal kit subscriber numbers.

- (c) (2 points) Explain how the sustainability review’s quotation challenges the newspaper’s claim.